

# BRASEN XU

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## EDUCATION

University of Waterloo, Bachelor of Computer Science (Co-op), AI Specialization  
3.8 GPA, 87.5 CAV

Sep 2022 - Apr 2027

## EXPERIENCE

### Splunk

May 2025 - August 2025

Backend Software Engineer Intern

San Francisco, CA

- Designed and integrated scalable **REST APIs** and UI workflows in **Java Spring Boot** for on-call incident response workflows, reducing investigation time by **30x** on average.
- Built **Java Spring microservices** enabling runtime configurations of **AWS EMR ML** pipelines, eliminating 6-hour redeployment cycles for parameter changes (**99% reduction**).
- Resolved flaky integration tests by fixing race conditions and enhancing **Mockito/JUnit** mocks, reducing CI/CD pipeline runtime by **75%** (7 hours to 1-2 hours).
- Developed self-service **AWS Glue** job execution API endpoints in **Java Spring**, cutting operational time by **93%** (30 mins to 2 mins) and removing manual console workflows.
- Awarded **Most Impactful Intern Project** in Security Org for achieving the term's top efficiency gains.

### Onsemi

September 2024 - December 2024

Software Engineer Intern

Waterloo, ON

- Designed and implemented new user interfaces for **5+** semiconductor chips using the **QT framework** and **C++**, enabling intuitive parameter tuning and real-time performance monitoring for manufacturing operators.
- Migrated legacy **Python 2** processing core to **Python 3**, improving execution speed by **20%** and enabling integration with modern frameworks.

### Siemens

January 2024 - April 2024

Software Engineer Intern

Concord, ON

- Developed **C# Windows Presentation Foundation (WPF)** display application with **SQL Server** backend for real-time parts tracking, reducing inventory lookup time by **96%** on average.
- Built a **Windows Form application** streamlining PCBA assembly workflows across **11 product lines**, reducing overall assembly errors by **30%**.

### Insurance Bureau of Canada

May 2023 - August 2023

Software Engineer Intern

Toronto, ON

- Created automated test scripts for manual test cases using **Selenium** and **Katalon Studio**, successfully boosting efficiency and reducing overall testing times by **over 95% (20x)**.

## PROJECTS

### MotiSpectra - ConUHacks Overall 2nd Place

- Built a real-time emotion and attentiveness analysis app using **Next.js**, **Python**, and custom **ML models**, integrating into Zoom/Meet/Teams and visualizing results via dynamic radar graphs with **97%** emotion detection accuracy.

### NarratorRL - Cohere Challenge 1st Place, MetHacks Overall 2nd Place

- Created a **React Native** app for vision-impaired users that uses **Tesseract OCR**, **Django**, and **Cohere NLP** to read text aloud, summarize content, detect language, and verify legibility in real time.

## SKILLS

|              |                                                                                     |
|--------------|-------------------------------------------------------------------------------------|
| Languages    | Java, Python, C++, JavaScript, TypeScript, C#, SQL, Bash, C, Kotlin, R, HTML        |
| Frameworks   | Spring Boot, React, Next.js, Node.js, Django, .NET, QT, Ruby on Rails, React Native |
| Tools        | AWS, Git, PostgreSQL, MongoDB, Kubernetes, Linux, Docker, Azure, Jira, Jupyter      |
| Technologies | REST APIs, Microservices, CI/CD, Machine Learning, Computer Vision, JUnit, Mockito  |